ETALE COVERS OF A CLOSED SUBSCHEME OF AN AFFINE NORMAL SCHEME WITH SEPARABLY CLOSED FUNCTION FIELD SPLIT INTO OPEN PIECES, AND FINITE SURJECTIVE ONES HAVE SECTIONS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the lift to an affine etale cover of X, the application of the structure lemma, the decomposition into open immersions that are also closed, and the displayed open-and-closed decomposition producing a section.

Editorial note. The label lemma-closed-of-affine-normal-scheme-with-alg-closed-function-field names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Lifting an etale morphism from a closed subscheme to the ambient scheme. Algebra, the lemma allowing an etale morphism V -> Z from a closed subscheme to be lifted to an affine scheme U etale over X. Supplies the lift U -> X to which the normal-scheme structure statement is applied.
- Separated etale morphisms over a normal scheme with separably closed function field. Same chapter, the lemma showing that on such a scheme a separated etale morphism is a disjoint union of open immersions. Supplies the first statement, applied to U -> X.

Source reference. Original lemma 04D1: sources/stacks-chow/tags.txt, source lines 5711–5711. Prerequisite pointer invoked at proof line(s) 14399; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09Z9: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 14312–14312. Prerequisite pointer invoked at proof line(s) 14401; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be an affine integral normal scheme with separably closed function field. Let $Z \subset X$ be a closed subscheme. Let $V \to Z$ be an étale morphism with V affine. Then V is a finite disjoint union of open subschemes of Z. If $V \to Z$ is surjective and finite étale, then $V \to Z$ has a section.*

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Proof. By Algebra, Lemma 04D1 we can lift V to an affine scheme U étale over X. Apply Lemma 09Z9 to $U \to X$ to get the first statement.

The final statement is a consequence of the first. Let $V = \coprod_{i=1,\ldots,n} V_i$ be a finite decomposition into open and closed subschemes with $V_i \to Z$ an open immersion. As $V \to Z$ is finite we see that $V_i \to Z$ is also closed. Let $U_i \subset Z$ be the image. Then we have a decomposition into open and closed subschemes

$$Z = \coprod_{(A,B)} \bigcap_{i \in A} U_i \cap \bigcap_{i \in B} U_i^c$$

where the disjoint union is over $\{1, \ldots, n\} = A \amalg B$ where A has at least one element. Each of the strata is contained in a single U_i and we find our section. $\square$